



**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE	:	INFORMATION SYSTEMS ENGINEERING
COURSE CODE	:	ISP550 / ITS572
EXAMINATION	:	JANUARY 2024
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts : PART A (10 Questions)
PART B (4 Questions)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
 - i) the Question paper
 - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

PART A

State whether each of the following statement is **TRUE** or **FALSE**.

1. Wisdom is a combination of information with necessary data and judgement of historical patterns.
2. In Scrum framework, each team consists of three roles: product owner, scrum master and stakeholder.
3. In System Development Life Cycle (SDLC), predictive approach is more flexible as compared to adaptive approach.
4. Vendor support is one of the criteria to consider when choosing off-the-shelf software.
5. Functional requirements specify criteria that can be used to judge the operation of a system, rather than specific behavior.
6. System test engineers use the documented requirements to understand what system is to be developed.
7. In domain class diagram, multiplicity is an attribute that describes a class.
8. In cloud computing service, the host neither owns the computers nor manages the configuration.
9. A situation that must be true before the use case begins is known as post-condition.
10. Phased installation occurs when old and new systems operated together for extended period.

(10 marks)

PART B**QUESTION 1**

"In a general sense, the term information system (IS) refers to a system of people, data records and activities that process the data and information in an organization, and it includes the organization's manual and automated processes." – Ray J. Paul, Journal of Computing and Information Technology - CIT 18, 2010, 2, 95–99

- a) Describe **THREE (3)** factors that promote the development of information systems in organizations.
(6 marks)
- b) Stakeholders are the people who have an interest in the successful implementation of a system. Differentiate between internal and external stakeholders.
(6 marks)
- c) Every semester, each student is required to register for courses through a system. Once registration is completed and validated, a slip will be produced.
What type of information system is used by the students to accomplish the registration activity? State a reason for your answer.
(4 marks)

QUESTION 2

Software development projects use different types of software development life cycle (SDLC) methodologies, depending on their nature and requirements, which basically define the way that the software development work is organized. The two main approaches are the waterfall method and the agile software development method.

- a) Describe **TWO (2)** major differences between both approaches.
(8 marks)
- b) You have been appointed to lead a development project where the requirements and needs are uncertain. Explain which approach you would choose for the project.
(6 marks)
- c) Scrum framework uses all the core principles of agile to define methods in facilitating a project.
 - i) Describe **THREE (3)** primary artefacts in the scrum process.
(6 marks)
 - ii) Explain the objective of daily stand-up meeting in agile team.
(4 marks)
- d) The process of project selection decisions must consider numerous factors and may have numerous outcomes. Describe **THREE (3)** possible decision outcomes.
(6 marks)

QUESTION 3

Systems design is the process of defining the architecture, modules, interfaces, and data for a system to satisfy specified requirements. Successively, implementation phase will come about.

- a) Describe the inputs and outputs of systems design phase. (4 marks)
- b) In a situation where clients refuse to own or administer their own hardware and software, explain other alternatives that you (as developer) may suggest to them. (6 marks)
- c) Systems interface and user interface design are among the activities involved in design phase.
 - i) What are the references used by interface designers to design user interface? (4 marks)
 - ii) Why should a designer consider systems interface? (4 marks)
- d) Sado Gym Inc. owns and operates fitness centres in several locations in Klang Valley. The centres have done well, and the company is planning an expansion by opening a new supercenter in the northern region. Sado Gym President, Abu Bakar, hired an IT consultant, Umar, to help develop an information system for the new facility. Umar has completed work on system architecture issues and his system specification design was approved. He is now ready to address system implementation tasks.

Suggest a changeover method for the new billing system by providing specific reasons to support your choice. Explain each step required for suggested changeover method.

(8 marks)

QUESTION 4

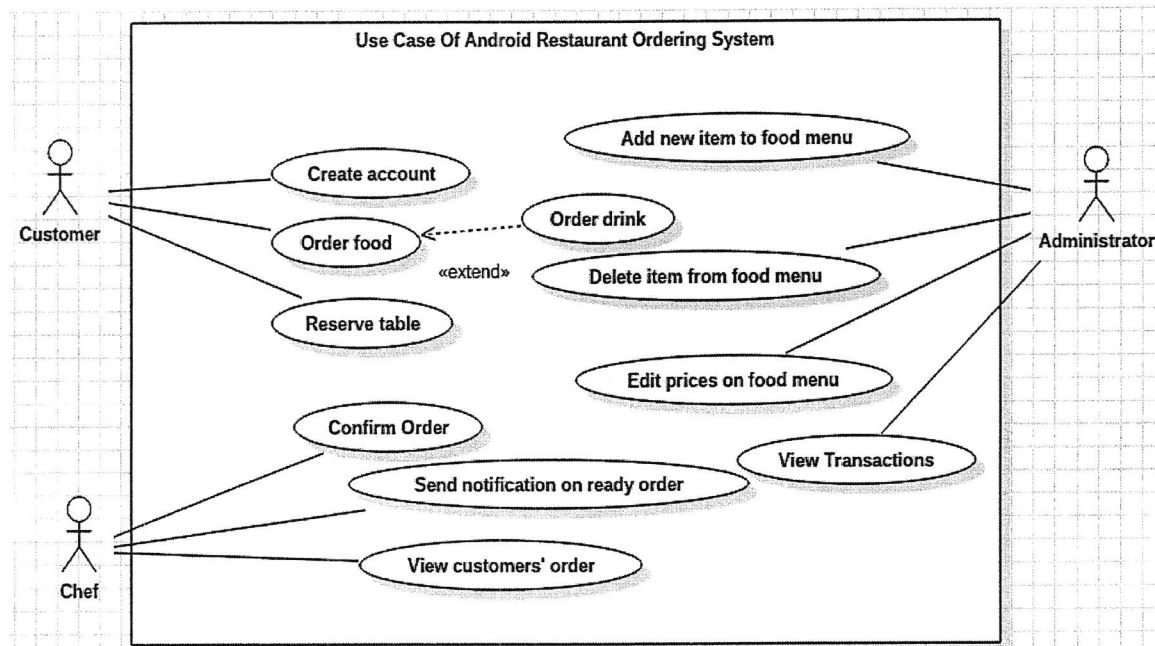


Figure 1

- a) Given the use case diagram in Figure 1, there are three actors with corresponding use cases. Customers are required to create an account prior to place order. If customers prefer to dine-in, they need to reserve a table. In order to reserve a table, customers must have ordered food (and drinks), provide date & time for the reservation as well as number of guests who intend to turn up for the reservation. However, a customer can only be accompanied by a maximum number of 10 guests. Once completed, a reservation number will be given to the customer.

You are required to complete the given use case description for use case Reserve Table by filling in the descriptions of (i) **Preconditions**, (ii) **Postconditions** and (iii) **Exception**.

(4 marks)

- b) Based on the Normal Flow description from Question 4 (a), construct a corresponding activity diagram.

(4 marks)

- c) State **TWO (2)** relevant fact-finding techniques that were possibly used by analysts so as to construct the given use case diagram.

(2 marks)

- d) Construct an appropriate domain class diagram for the proposed system.

(8 marks)

Use Case Name:	Reserve table	
Actors:	Customer	
Description:	Customer who prefers to dine-in has to make table reservation.	
Trigger:	User triggered	
Preconditions:	(i)	
Postconditions:	(ii)	
Normal Flow:	Actor	System
	1. Customer chooses to make reservation 2. Customer enters date, time and number of patrons	3. System checks the availability of table. 4. System displays reservation number
Alternative Flows:	Customer cancels to make reservation. The use case ends.	
Exceptions:	(iii)	
Includes:		
Business Rules:	A customer can be accompanied by a maximum of 10 guests.	
Assumptions:		

END OF QUESTION PAPER